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Inventor(s)	Kinoshita; Hideaki et al.

Catheter assembly and medical valve

Abstract

A catheter assembly is provided with a catheter, a catheter hub, and a valve body (medical valve) provided in the catheter hub. The valve body includes a hollow main body provided with a distal end surface and a fixing portion for fixing the valve body to the catheter hub. At least a part of an outer peripheral surface and at least a part of an inner peripheral surface of the main body are inclined with respect to a central axis of the valve body. The main body of the valve body includes a distal end slit provided on the distal end surface and a side slit provided on the outer peripheral surface of the main body and continuously extending from the distal end slit.

Inventors:	Kinoshita; Hideaki (Kofu, JP), Yamamoto; Shota (Kai, JP)
Applicant:	TERUMO KABUSHIKI KAISHA (Tokyo, JP)
Family ID:	1000008820532
Assignee:	TERUMO KABUSHIKI KAISHA (Tokyo, JP)
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Assistant Examiner: Allen; Robert F

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of and claims benefit to PCT Application No. PCT/JP2018/036190, filed on Sep. 28, 2018, entitled “CATHETER ASSEMBLY AND MEDICAL VALVE” which claims priority to Japanese Patent Application No. 2017-191169, filed on Sep. 29, 2017. The entire disclosures of the applications listed above are hereby incorporated herein by reference, in their entirety, for all that they teach and for all purposes.

FIELD

(1) The present disclosure relates to catheter assemblies, including catheters, with which a blood vessel is punctured for transfusion and the like to a patient. The present disclosure also relates to medical valves.

BACKGROUND

(2) Conventionally, for example, a catheter assembly is used to perform transfusion or the like to a patient. This type of catheter assembly is provided with a catheter, a catheter hub fixed to a proximal end of the catheter, an inner needle inserted into the catheter, and a needle hub fixed to a proximal end of the inner needle. This type of catheter assembly includes one in which a valve is provided in the catheter hub in order to prevent a blood backflow (outflow of blood from the catheter hub) when the catheter is inserted into the body. As one mode of the valve, there is a so-called disc valve in which a slit is provided in a disc-shaped valve body. In a case of the disc valve, it is necessary to open the valve body by advancing a separate component referred to as a plug arranged in the catheter hub at the time of transfusion.

(3) In contrast, there also is a type of assembly in which a duckbill valve (one-way valve) is provided in the catheter hub for the similar purpose (for example, refer to JP 2-164376 A). In a case of the duckbill valve, when blood flows into the catheter hub through the catheter, a blood pressure in a proximal direction is applied to the valve body to deform the same, so that the valve body is closed. On the contrary, when a drug solution is supplied from a proximal end side of the catheter

hub, a pressure of the drug solution is applied to the valve body in a distal direction to deform the same, so that the valve body is opened. Therefore, in a case of the duckbill valve, the separate component for opening is not necessary.

SUMMARY

(4) In a case of the conventional duckbill valve, the separate component for opening is not required, but there is a disadvantage that a flow rate of the drug solution is likely to be reduced because of a structure to open by using the pressure of the drug solution. It is desired that the valve body may be opened more easily even in a case where the separate component for opening is used.

(5) A catheter assembly according to the present disclosure is provided with a catheter, a catheter hub fixed to the catheter, and a valve body provided in the catheter hub, in which the valve body includes a hollow main body provided with a distal end surface located at a distal end of the valve body and a fixing portion for fixing the valve body to the catheter hub, at least a part of an outer peripheral surface of the main body is inclined with respect to a central axis of the valve body, at least a part of an inner peripheral surface in an internal space of the main body is inclined with respect to the central axis, and the main body includes a distal end slit provided on the distal end surface and a side slit provided on the outer peripheral surface of the main body and continuously extending from the distal end slit.

(6) In one embodiment of the present disclosure, configured as described above, the main body of the valve body is provided with not only the distal end slit but also the side slit continuously extending from the distal end slit, so that the main body is easily deformed when the pressure (e.g., blood pressure or drug solution pressure, etc.) is applied to the main body. Therefore, when the blood pressure is applied to the main body from a distal end side, the main body is deformed inward and the distal end slit and the side slit close, so that a blood backflow may be suppressed. In contrast, when the drug solution pressure is applied to the main body from a proximal end side, the main body is deformed outward and not only the distal end slit but also the side slit opens, so that a flow rate at the time of opening may be increased. Therefore, according to the present disclosure, it is not necessary to use a separate component for opening the valve, and it is possible to increase the flow rate at the time of opening compared to the flow rate of a catheter assembly with a structure using a conventional duckbill valve.

(7) A pair of side slits may be continuous from both ends in a slit length direction of the distal end slit.

(8) With this configuration, when the pressure is applied from the distal end side or the proximal end side of the valve body, the main body is more easily deformed inward or outward. Therefore, a function as a check valve (e.g., a function of stopping or suppressing blood and/or a function of allowing a drug solution to flow, etc.) may be improved.

(9) The valve body may include a cylindrical portion formed to be thinner than the fixing portion between the main body and the fixing portion. The cylindrical portion may be disposed between the main body and the fixing portion, or base. In one embodiment, the cylindrical portion may have an outer diameter that is less than an outer diameter of the fixing portion.

(10) With this configuration, a proximal end of the main body is not directly connected to a distal end of the fixing portion but is connected to a thinner cylindrical portion, so that the main body is more easily deformed when the pressure is applied to the main body.

(11) A proximal end of the side slit may be located at a proximal end of the cylindrical portion.

(12) With this configuration, the main body is more easily deformed, so that the function as the check valve (e.g., the function of stopping or suppressing the blood and/or the function of allowing the drug solution to flow) may be improved.

(13) The proximal end of the side slit may be located on a distal end side with respect to the proximal end of the main body.

(14) The proximal end of the side slit may be located on a distal end side with respect to the distal end of the fixing portion.

- (15) The proximal end of the side slit may be located on a proximal end side with respect to a central position in an axial direction of the main body.
- (16) The proximal end of the side slit may be located on a distal end side with respect to the central position in the axial direction of the main body.
- (17) The catheter assembly may be provided with an inner needle inserting into the catheter, and in a state in which the inner needle inserts into, and/or through, the valve body, a gap may be formed between slit surfaces forming the distal end slit and an outer surface of the inner needle.
- (18) With this configuration, at the time of the puncture of the blood vessel with the distal end of the catheter assembly, it is possible to allow air to flow from the distal end side to the proximal end side of the valve body through the gap formed between the slit surfaces and the inner needle. As a result, the catheter hub may be vented before the inner needle is removed from the catheter, and the distal end side with respect to the valve body in the catheter hub may be filled with the blood.
- (19) A hollow opening member arranged in a lumen of the catheter hub which is displaceable in the distal direction with respect to the catheter hub to open the valve body may be provided.
- (20) With this configuration, it is possible to open the valve body even when a moving distance from when the opening member starts pushing the valve body is shorter than that in a case of a combination of a disc-shaped valve body (so-called disc valve) and the opening member. The valve body may be opened with a smaller force than that in a case of the combination of the disc-shaped valve body and the opening member, so that the operation is easy.
- (21) In a state in which the opening member opens the valve body, the distal end slit of the valve body may be deformed along an outer shape of the opening member, and a gap of the side slit may be widened in a distal direction.
- (22) A medical valve according to the present disclosure is provided with a hollow main body including a distal end surface, and a base located on a proximal end side of the main body, in which at least a part of an outer peripheral surface of the main body is inclined with respect to a central axis of the main body, at least a part of an inner peripheral surface in an internal space of the main body is inclined with respect to the central axis, and the main body includes a distal end slit provided on the distal end surface and a side slit provided on the outer peripheral surface of the main body and continuously extending from the distal end slit.
- (23) According to the catheter assembly and medical valve of the present disclosure, it is not necessary to use a separate component for opening the valve, and it is possible to increase the flow rate at the time of opening from that with a structure using a conventional duckbill valve.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a catheter assembly according to an embodiment of the present disclosure;
- (2) FIG. 2 is a partial cross-sectional view of the catheter assembly taken along line II-II in FIG. 1;
- (3) FIG. 3 is a perspective view of a valve body in accordance with embodiments of the present disclosure;
- (4) FIG. 4 is a cross-sectional view of the valve body shown in FIG. 3;
- (5) FIG. 5A is a cross-sectional view of a valve body illustrating an example in which a length of a side slit is made shorter than the slit of the valve body shown in FIG. 4;
- (6) FIG. 5B is a cross-sectional view of a valve body illustrating an example in which a length of the side slit is made further shorter than the slit of the valve body shown in FIG. 5A;
- (7) FIG. 6A is a perspective view illustrating the valve body in a state in which an inner needle inserts through the same;
- (8) FIG. 6B is a cross-sectional view taken along line VIB-VIB in FIG. 6A;

- (9) FIG. 7 is a partial cross-sectional view illustrating a closure state of the valve body in accordance with embodiments of the present disclosure;
- (10) FIG. 8 is a partial cross-sectional view illustrating an opening state of the valve body in accordance with embodiments of the present disclosure;
- (11) FIG. 9 is a perspective view of a valve body according to one embodiment of the present disclosure;
- (12) FIG. 10 is a cross-sectional view of a catheter member before opening a valve body according to an embodiment of the present disclosure; and
- (13) FIG. 11 is a cross-sectional view of the catheter member shown in FIG. 10 after opening the valve body.

DETAILED DESCRIPTION

- (14) Embodiments of a catheter assembly and a medical valve according to the present disclosure are hereinafter described with reference to the accompanying drawings.
- (15) Referring to FIG. 1, a catheter assembly 10 is illustrated comprising a catheter 12, a catheter hub 14 connected to a proximal end side of the catheter 12, an inner needle 16 including a sharp needle tip 16a at a distal end thereof, the inner needle 16 inserting into the catheter 12, a needle hub 18 connected to the inner needle 16, and a valve body 20 (e.g., a medical valve, etc.) arranged in the catheter hub 14.
- (16) The catheter assembly 10 is arranged such that the needle hub 18 can be gripped by a user (e.g., a doctor, nurse, medical worker, or the like), and a blood vessel of a patient is punctured with a distal end thereof. In an initial state before use (e.g., before puncture of the patient), the catheter assembly 10 has a double tube structure in which the inner needle 16 inserts into the catheter 12, and the inner needle 16 protrudes from a distal end of the catheter 12 by a predetermined length.
- (17) The catheter 12 is a small-diameter tubular member having flexibility. The catheter 12 may be formed of, for example, a resin material, preferably a soft resin material. A hollow catheter hub 14 is connected to be fixed to a proximal end of the catheter 12. The catheter 12, the catheter hub 14, and the valve body 20 form a catheter member 22. As illustrated in FIG. 2, the proximal end of the catheter 12 is fixed to a distal end of the catheter hub 14 by a fixing member 24 arranged in the distal end of the catheter hub 14.
- (18) When the catheter assembly 10 is used, the catheter hub 14 is exposed on a skin of the patient in a state in which the blood vessel is punctured with the catheter 12, and is adhered to the skin with a dressing material, a tape or the like to be indwelled thereon.
- (19) In FIG. 1, the inner needle 16 is a tubular member having rigidity capable of puncturing the skin of the patient, and is formed of, for example, a metal material (e.g., stainless steel or the like). On a top surface of a distal end of the inner needle 16, a groove 26 for checking blood flashback is formed. In the initial state of the catheter assembly 10 illustrated in FIG. 1, a proximal end of the inner needle 16 is arranged in a hollow portion of the catheter hub 14. Note that the inner needle 16 may also be formed as a solid needle.
- (20) The needle hub 18 is connected to the proximal end of the inner needle 16. The needle hub 18 includes a hub main body 28 which serves as a grip that is gripped by the user and a needle holder 30 which protrudes from the hub main body 28 in a distal direction and holds the proximal end of the inner needle 16. In the initial state of the catheter assembly 10, the needle holder 30 is inserted into a proximal end of the catheter hub 14.
- (21) The valve body 20 may correspond to a check valve (e.g., a one-way valve) which prevents a backflow of blood in a proximal direction (outflow of blood from the catheter hub 14) and allows passage of a drug solution in the distal direction. A material forming the valve body 20 may include, for example, a synthetic rubber such as polybutadiene series, nitrile series, and chloroprene series, a natural rubber such as polyisoprene, or a thermosetting elastomer such as urethane rubber, silicon rubber, and fluorine rubber, thermoplastic elastomer, or other elastomers.
- (22) As illustrated in FIG. 2, the valve body 20 is provided with a hollow main body 32 (e.g., the

bill portion of the valve body 20) provided with a distal end surface 32a located at a distal end of the valve body 20, a fixing portion 34 provided on a proximal side with respect to the main body 32 for fixing the valve body 20 to the catheter hub 14, and a cylindrical portion 36 provided between the main body 32 and the fixing portion 34. The main body 32 and the cylindrical portion 36 protrude from the fixing portion 34 in the distal direction, that is, toward the catheter 12. In the initial state of the catheter assembly 10, the inner needle 16 penetrates the main body 32 in an axial direction. The cylindrical portion 36 may be disposed between the main body 32 and the fixing portion 34, or base. In one embodiment, the cylindrical portion 36 may have an outer diameter that is less than an outer diameter of the fixing portion 34.

(23) At least a part of an outer peripheral surface 33 of the main body 32 is inclined with respect to a central axis a of the valve body 20. At least a part of an inner peripheral surface in an internal space 32s of the main body 32 is inclined with respect to the central axis a. Specifically, the main body 32 includes a pair of inclined portions 38 formed on opposite sides with respect to the central axis a of the valve body 20. A distal end wall portion 32w (refer also to FIG. 4) is provided at a distal end of the pair of inclined portions 38.

(24) Each inclined portion 38 includes an inclined outer surface 40 and an inclined inner surface 42 inclined so as to approach the central axis a of the valve body 20 in the distal direction. Therefore, the main body 32 includes a pair of inclined outer surfaces 40 and a pair of inclined inner surfaces 42. The inclined outer surface 40 forms a part of the outer peripheral surface 33 of the main body 32. The inclined inner surface 42 forms a part of the inner peripheral surface of the main body 32. The inclined outer surface 40 and the inclined inner surface 42 are flat surfaces in a natural state (e.g., in a state in which no external force acts upon the main body 32).

(25) FIG. 2 also illustrates various additional features described in more detail below. In more detail, FIG. 2 shows that in a state where the valve body 20 is fixed in the catheter hub 14, an entirety of the (hollow) main body 32 and an entirety of the cylindrical portion 36 are spaced apart from an inner peripheral surface 14s (at 14s1) of the catheter hub 14. In addition, the inner peripheral surface 14s of the catheter hub 14 comprises a second groove 72 that engages with the fixing portion 34 to fix the valve body 20 in the catheter hub 14. In the state where the valve body 20 is fixed in the catheter hub 14, a first surface 74 of the second groove 72 contacts a first surface 76 of the fixing portion 34. In the state where the valve body 20 is fixed in the catheter hub 14, a second surface 78 of the second groove 72 connected to the first surface 74 of the second groove 72 contacts a second surface 80 of the fixing portion 34. In the state where the valve body 20 is fixed in the catheter hub 14, the first surface 74 of the second groove 72 is closer to the distal end of the valve body 20 than the second surface 78 of the second groove 72.

(26) As shown, the fixing portion 34 comprises a third surface 82 that is exposed to a lumen 84 of the catheter hub 14. As also shown, the second surface 80 of the fixing portion 34 connects the first and third surfaces 76 and 82 of the fixing portion 34. Further, the inner peripheral surface 14s of the catheter hub 14 comprises a first inclined portion 86 extending from an end of the second surface 78 of the second groove 72 and that is exposed to the lumen 84 of the catheter hub 14. As shown, the first inclined portion 86 is inclined in a direction away from the third surface 82 of the fixing portion 34 and toward the central axis a of the valve body 20 when moving in a first direction that is away from the distal end of the valve body 20. In the example shown, the first inclined portion 86 extending from the end of the second surface 78 of the second groove 72 is inclined at a different angle with respect to the central axis a of the valve body 20 compared to the first and second surfaces 74 and 78 of the second groove 72. As further shown, the inner peripheral surface 14s of the catheter hub 14 comprises a second inclined portion 88 extending from an end of the first inclined portion 86. The second inclined portion 88 is inclined away from the central axis a of the valve body 20 when moving in the first direction.

(27) As illustrated in FIG. 3, the inclined outer surface 40 has a width which decreases in the proximal direction. The outer peripheral surface 33 of the main body 32 includes a cylindrical

surface **33a**. In this embodiment, the distal end surface **32a** of the main body **32** is a flat surface formed into a substantially rectangular shape and is perpendicular to the central axis **a**. A short side of the distal end surface **32a** is continuous to a distal end of the cylindrical surface **33a**. A long side of the distal end surface **32a** is continuous to a distal end of the inclined outer surface **40**.

(28) The fixing portion **34** is formed into a circular ring shape. An outer diameter of the fixing portion **34** is larger than a maximum outer diameter of the main body **32**. Therefore, the fixing portion **34** has a shape protruding radially outward at the proximal end of the valve body **20**. As illustrated in FIG. 2, the fixing portion **34** engages with a fixing groove **44** provided on an inner peripheral surface **14s** of the catheter hub **14**, so that the valve body **20** is fixed in a predetermined position in the catheter hub **14**. In FIG. 2, the main body **32** and the cylindrical portion **36** are separated from an inner peripheral surface **14s1** on the distal end side with respect to the fixing portion **34** out of the inner peripheral surface **14s** of the catheter hub **14** over an entire circumference.

(29) The cylindrical portion **36** forms a portion between the main body **32** and the fixing portion **34**. The cylindrical portion **36** and the fixing portion **34** form a base of the valve body **20**. The cylindrical portion **36** is formed to be thinner than the fixing portion **34**. Specifically, an outer diameter of the cylindrical portion **36** is constant in the axial direction and is smaller than the outer diameter of the fixing portion **34**. The outer diameter of the cylindrical portion **36** is the same as the maximum outer diameter of the main body **32**. An inner diameter of the cylindrical portion **36** is constant in the axial direction and is the same as an inner diameter of the fixing portion **34**. Note that the inner diameter of the cylindrical portion **36** may be different from the inner diameter of the fixing portion **34**.

(30) As illustrated in FIG. 3, the main body **32** includes a distal end slit **46** provided on the distal end surface **32a** and a side slit **48** provided on the outer peripheral surface **33** of the main body **32** and continuously extending from the distal end slit **46**. The distal end slit **46** extends in a direction orthogonal to the central axis **a** of the valve body **20** and both ends thereof reach the outer peripheral surface **33** of the main body **32**. As illustrated in FIG. 4, the distal end slit **46** reaches a proximal end surface of the distal end wall portion **32w** of the main body **32**. Therefore, a proximal end **46e** of the distal end slit **46** faces the internal space **32s** of the main body **32**.

(31) As illustrated in FIG. 3, the side slit **48** extends in the proximal direction along the outer peripheral surface **33** of the main body **32** from both ends of the distal end slit **46**. Therefore, a pair of side slits **48** are provided. Each side slit **48** reaches the inner peripheral surface of the main body **32** (e.g., faces the internal space **32s**).

(32) In this embodiment, the pair of side slits **48** extend from both the ends of the distal end slit **46** so as to be orthogonal thereto in parallel with each other with the same length. Note that, depending on a shape of the main body **32**, the pair of side slits **48** may extend so as to be non-parallel to each other. The pair of side slits **48** may extend so as to be non-parallel to the central axis **a** of the valve body **20**. The pair of side slits **48** may be formed with different lengths.

(33) As illustrated in FIG. 4, a proximal end **48e** of the side slit **48** is located at a proximal end of the cylindrical portion **36** (e.g., at the distal end of the fixing portion **34**). Therefore, in this embodiment, a total length **Ls** of the side slit **48** in the axial direction of the valve body **20** is longer than a total length **Lb** of the main body **32** and shorter than a total length **Lh** of the valve body **20**.

(34) As illustrated in FIGS. 5A and 5B, the proximal end **48e** of the side slit **48** may also be located on the distal end side with respect to the distal end of the fixing portion **34**. In FIGS. 5A and 5B, the proximal end **48e** of the side slit **48** is located on the distal end side with respect to a proximal end of the main body **32** (e.g., the distal end of the cylindrical portion **36**). More specifically, in FIG. 5A, the proximal end **48e** of the side slit **48** is located on the proximal end side with respect to the central position in the axial direction of the main body **32** (e.g., between the central position in the axial direction of the main body **32** and the proximal end of the main body **32**). In FIG. 5B, the proximal end **48e** of the side slit **48** is located on the distal end side with respect to the central

position in the axial direction of the main body **32** (between the proximal end surface of the distal end wall portion **32w** and the central position in the axial direction of the main body **32**).

(35) As illustrated in FIGS. **6A** and **6B**, in a state in which the inner needle **16** penetrates the valve body **20** (e.g., in an initial state of the catheter assembly **10**), a site of the distal end wall portion **32w** of the valve body **20** which the inner needle **16** penetrates curves with elastic deformation. That is, the distal end wall portion **32w** is pushed outward by the inner needle **16**. Therefore, a triangular gap **G** is formed between slit surfaces **46s** which form the distal end slit **46** and an outer peripheral surface of the inner needle **16**. The gap **G** is formed in two sites opposite to each other across the inner needle **16**. The gap **G** is formed so that gas (air) may pass therethrough. The gap **G** may be set to a size such that liquid (blood) passes therethrough with difficulty. Note that, in the state in which the inner needle **16** inserts into the valve body **20**, it is possible that no gap is formed between the slit surfaces **46s** and the outer peripheral surface of the inner needle **16** (e.g., the slit surface **46s** may come into contact with the outer peripheral surface of the inner needle **16** over an entire circumference of the inner needle **16**). The gap **G** may extend over an entire distal end slit **46**. Furthermore, the gap **G** may extend to a distal end of the side slit **48**.

(36) Next, an action of the catheter assembly **10** configured as described above is described.

(37) When the catheter assembly **10** is used, the user (doctor, nurse or the like) grips the needle hub **18** of the catheter assembly **10** in the initial state illustrated in FIG. **1** and punctures the blood vessel of the patient with the distal end portion of the catheter assembly **10** (e.g., the distal ends of the inner needle **16** and the catheter **12**). With the puncture, the blood of the patient flows into the groove **26** formed at the distal end of the inner needle **16** (e.g., where flashback occurs). As a result, it is confirmed that the distal end of the catheter **12** secures the blood vessel. After the puncture, by pushing the catheter hub **14** in the distal direction while maintaining the positions of the needle hub **18** and the inner needle **16**, the catheter **12** is advanced to be inserted into the blood vessel.

(38) After the catheter **12** is inserted into the blood vessel by a predetermined length, the needle hub **18** is next pulled in the proximal direction in a state in which the position of the catheter member **22** is maintained, and the inner needle **16** is removed from the catheter member **22**. At that time, the inner needle **16** is removed from the valve body **20**. As a result, the catheter member **22** is indwelled in the patient side. After removing the inner needle **16** from the catheter member **22**, the catheter hub **14** is fixed to the patient with the dressing material, tape or the like.

(39) Since the valve body **20** is provided in the catheter hub **14**, the blood outflow from the catheter hub **14** is prevented even after the inner needle **16** is removed from the valve body **20**. That is, as illustrated in FIG. **7**, when the blood flows into the catheter hub **14** via the catheter **12** and a blood pressure is applied to the main body **32** of the valve body **20** from the distal end side of the valve body **20**, the main body **32** (inclined portion **38**) receives an inward force. As a result, the valve body **20** is closed, and the flow of the blood from the distal end side to the proximal end side of the valve body **20** is prevented or suppressed.

(40) Next, a connector of a transfusion tube not illustrated is connected to a proximal end side of the catheter member **22** in a state in which the inner needle **16** is removed (e.g., from the proximal end of the catheter hub **14**), and the drug solution (e.g., a transfusion material) is administered from the transfusion tube to the patient.

(41) When the drug solution is supplied from a proximal end side of the catheter hub **14** into the catheter hub **14**, a drug solution pressure is applied to the main body **32** of the valve body **20** from the proximal end side of the valve body **20** as illustrated in FIG. **8**. Then, the main body **32** (e.g., the inclined portion **38**) receives an outward force. As a result, the valve body **20** is opened, and the flow of the drug solution from the proximal end side to the distal end side of the valve body **20** is allowed.

(42) In this case, the catheter assembly **10** according to this embodiment has the following non-limiting effects.

(43) According to the catheter assembly **10**, as illustrated in FIG. **3**, the main body **32** of the valve

body **20** is provided with not only the distal end slit **46** but also the side slit **48** continuously extending from the distal end slit **46**, so that the main body **32** is easily deformed when the pressure (e.g., via blood pressure and/or drug solution pressure, etc.) is applied to the main body **32**. Therefore, as illustrated in FIG. 7, when the blood pressure is applied to the main body **32** from the distal end side, the main body **32** is deformed inward and the distal end slit **46** and the side slit **48** close, so that a blood backflow may be suppressed or prevented. In contrast, as illustrated in FIG. 8, when the drug solution pressure is applied to the main body **32** from the proximal end side, the main body **32** is deformed outward and not only the distal end slit **46** but also the side slit **48** opens, so that a flow rate at the time of opening may be increased. Therefore, according to the present disclosure, it is not necessary to use a separate component (plug) for opening, and it is possible to increase the flow rate at the time of opening without requiring a structure that is typically used to open a conventional duckbill valve.

(44) Especially, in this embodiment, as illustrated in FIG. 3, the pair of side slits **48** are continuous from both the ends in a slit length direction of the distal end slit **46**. With this configuration, when the pressure is applied from the distal end side or the proximal end side of the valve body **20**, the main body **32** is more easily deformed inward or outward, respectively. Therefore, a function as the check valve (e.g., the function of stopping or suppressing the blood and function of allowing the drug solution to flow therethrough) for the valve body **20** may be further improved over conventional valves and assemblies.

(45) The valve body **20** includes the cylindrical portion **36** formed to be thinner than the fixing portion **34** between the main body **32** and the fixing portion **34**. With this configuration, the proximal end of the main body **32** is not directly connected to the distal end of the fixing portion **34** but is connected to a thinner cylindrical portion **36**, so that the main body **32** is more easily deformed when the pressure is applied to the main body **32**.

(46) As illustrated in FIG. 4, the proximal end **48e** of the side slit **48** is located at the proximal end of the cylindrical portion **36**. With this configuration, the main body **32** is more easily deformed, so that the function as the check valve (e.g., the function of stopping or suppressing the blood and function of allowing the drug solution to flow) may be further improved.

(47) As illustrated in FIGS. 6A and 6B, in a state in which the inner needle **16** inserts into the valve body **20**, the gap **G** is formed between the slit surfaces **46s** which form the distal end slit **46** and the outer peripheral surface of the inner needle **16**. With this configuration, at the time of the puncture of the blood vessel with the distal end of the catheter assembly **10**, it is possible to allow air to flow from the distal end side to the proximal end side of the valve body **20** through the gap **G** formed between the slit surfaces **46s** and the inner needle **16**. As a result, the catheter hub **14** may be vented before the inner needle **16** is removed from the catheter **12**, and the distal end side with respect to the valve body **20** in the catheter hub **14** may be filled with the blood.

(48) In the catheter assembly **10** described above, instead of the valve body **20**, a valve body **50** provided with a conical main body **52** illustrated in FIG. 9 may be used. A distal end slit **56** is provided on a distal end surface **52a** of the main body **52**. On an outer peripheral surface **53** (e.g., a conical surface) of the main body **52**, side slits **58** extending continuously from both ends of the distal end slit **56** are provided. Therefore, also in a case where this valve body **50** is used, the effect similar to that in a case where the above-described valve body **20** is used may be obtained.

(49) In the catheter assembly **10** described above, a catheter member **22a** illustrated in FIG. 10 may also be used. The catheter member **22a** is provided with an opening member **60** arranged in a lumen of the catheter hub **14**. The opening member **60** is a hollow plug arranged so as to be displaceable in the axial direction of the catheter hub **14** and is displaced in the distal direction with respect to the catheter hub **14** to open the valve body **20**. As illustrated in FIG. 10, before the displacement in the distal direction (initial position), the opening member **60** is arranged in a proximal end side space **S2** out of a distal end side space **S1** and the proximal end side space **S2** separated by the valve body **20** in the catheter hub **14**.

(50) The opening member **60** includes a hollow body portion **62** which forms a proximal end side of the opening member **60** and a hollow cylindrical pressing portion **64** which forms a distal end side of the opening member **60**. The body portion **62** is supported by the inner peripheral surface of the catheter hub **14** and is slidable in the axial direction with respect to the inner peripheral surface of the catheter hub **14**. The pressing portion **64** protrudes in the distal direction from a distal end of the body portion **62** and is formed to have a smaller diameter than that of the body portion **62**. A distal end of the pressing portion **64** is inserted into the valve body **20** in an initial position illustrated in FIG. **10**. A tapered portion **64t** of the opening member **60**, an outer diameter of which, decreases in the distal direction is provided at the distal end of the pressing portion **64**.

(51) An inner member **68** is fixed to the proximal end of the catheter hub **14**. The inner member **68** restricts displacement of the opening member **60** from the initial position in the proximal direction with respect to the catheter hub **14**. The inner member **68** is fitted to the lumen of the catheter hub **14** in a state of being located on the proximal end side with respect to the opening member **60** to be fixed with respect to the catheter hub **14**.

(52) When using the catheter assembly **10** in which the catheter member **22a** configured in this manner is adopted, as illustrated in FIG. **11**, after the inner needle **16** (FIG. **1**) is removed from the catheter member **22a**, the user attaches the connector **70** of the transfusion tube to a proximal end opening of the catheter hub **14** (inner member **68**). A distal end of the connector **70** comes into contact with the proximal end of the opening member **60** at the time of insertion and pushes out the opening member **60** in the distal direction. The opening member **60** penetrates the distal end slit **46** of the valve body **20** as this moves (advances) in the distal direction. That is, the pressing portion **64** of the opening member **60** which advances abuts the inclined inner surface **42** of the main body **32** of the valve body **20** and pushes out the main body **32** from the inside. As a result, the main body **32** opens with elastic deformation. The distal end slit **46** of the valve body **20** is deformed along an outer shape of the opening member **60**, and the gap of the side slit **48** is widened in the distal direction. Specifically, the side slit **48** of the pair of inclined portions **38** is deformed in an oblique direction with respect to an axis line from the proximal end **48e** toward the distal end surface **32a**. As a result, a lumen of the catheter **12**, the lumen of the catheter hub **14**, a lumen of the opening member **60**, and a supply path of the connector **70** may be communicated with one another to deliver the transfusion material into the blood vessel of the patient.

(53) According to a configuration obtained by combining the duckbill-type valve body **20** including the side slit **48** in addition to the distal end slit **46** and the opening member **60** just like the catheter member **22a**, it is possible to open the valve body **20** even when a moving distance from when the opening member **60** starts pushing the valve body **20** is shorter than that in a case of a combination of a disc-shaped valve body (so-called disc valve) and the opening member **60**. That is, since the valve body **20** is provided with the side slit **48**, the distal end slit **46** and the side slit **48** start opening simply when the opening member **60** (pressing portion **64**) pushes the main body **32** of the valve body **20** a little in the distal direction from the inside (the valve body **20** is half-open), so that the valve body **20** is opened quickly. According to the configuration of the combination of the valve body **20** and the opening member **60**, the valve body **20** may be opened with a smaller force than that in a case of the combination of the disc valve and the opening member **60**, so that the operation is easy. Note that, in the catheter member **22a**, the valve body **50** (FIG. **9**) may be used instead of the valve body **20**.

(54) The present disclosure is applicable to other medical valves arranged in a fluid passage of a medical device in addition to the valve bodies **20** and **50** for the catheter assembly **10** described above.

(55) The present disclosure is not limited to the above-described embodiments, and various modifications may be made without departing from the scope of the present disclosure.

Claims

1. A catheter assembly comprising: a catheter; a catheter hub fixed to the catheter; a valve body provided in the catheter hub; and an opening member and an inner member, wherein the valve body includes a hollow main body provided with a distal end surface located at a distal end of the valve body, and a fixing portion for fixing the valve body to the catheter hub, wherein at least a part of an outer peripheral surface of the hollow main body is inclined with respect to a central axis of the valve body, wherein at least a part of an inner peripheral surface in an internal cavity of the hollow main body is inclined with respect to the central axis, wherein the hollow main body includes a distal end slit provided on the distal end surface and a side slit provided on the outer peripheral surface of the hollow main body and continuously extending from the distal end slit, wherein the fixing portion comprises a proximal surface that is exposed to a lumen of the catheter hub in a state where the valve body is fixed in the catheter hub, wherein, in a cross-sectional view taken along a direction that is perpendicular to the distal end of the valve body, one side of an inner surface of the valve body has a first surface portion that connects a second surface portion that is parallel to the central axis of the valve body to a third surface portion that is perpendicular to the central axis of the valve body, wherein a first transition between the first surface portion and the second surface portion forms a first inward facing obtuse angle, wherein a second transition between the first surface portion and the third surface portion forms a second inward facing obtuse angle, wherein the second surface portion extends from the proximal surface of the fixing portion to the first surface portion, wherein the third surface portion of the inner surface of the valve body is opposite the distal end surface of the hollow main body, wherein the opening member comprises: a hollow cylindrical pressing portion which forms a distal side of the opening member; a hollow body portion extending proximally from the hollow cylindrical pressing portion; a proximal portion of the opening member which is supported by an inner peripheral surface of the catheter hub and is slidable in an axial direction with respect to the inner peripheral surface of the catheter hub; and a distal end of the hollow cylindrical pressing portion which is disposed in the internal cavity of the hollow main body of the valve body in an initial state in which no external force acts upon the opening member, and wherein the inner member comprises: a cylindrical body which is fitted to the inner peripheral surface of the catheter hub; an inner cavity into which a connector of an infusion tube can be attached; a distal portion; and a distal end surface that is arranged to abut against the proximal portion of the opening member in the initial state in which no external force acts upon the opening member, wherein the proximal portion of the opening member comprises an opening member proximal protruding portion which protrudes inwardly from the distal portion of the inner member in the initial state so that the opening member proximal protruding portion contacts a tip surface of the connector in a connector insertion state.
 2. The catheter assembly of claim 1, wherein: the valve body includes a cylindrical portion extending between the hollow main body and the fixing portion; the cylindrical portion has a constant outer diameter along the central axis of the valve body; the valve body has a pair of flat surfaces with no slits in a state in which no external force acts upon the main body, and each flat surface is between a pair of curved surfaces in a circumferential direction of the valve body; and the side slit terminates at a boundary between the fixing portion and the cylindrical portion, the side slit is disposed along the pair of curved surfaces parallel to the central axis of the valve body.
 3. The catheter assembly of claim 2, wherein: the pair of flat surfaces each comprise a proximal end; the side slit extends further than the proximal end of each flat surface along the central axis of the valve body.
 4. The catheter assembly of claim 1, wherein the inner member comprises a proximal end including a flange shaped end covering a proximal end of the catheter hub.
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